



Fig. 4: TP4056 charging module



Fig. 5: DC-DC booster module



Fig. 6: OTG mini fan

of Li-ion battery is 4.2 volts but, for safety reasons, charging stops at 4.1 volts.

The power bank's output can be used to drive an emergency light, charge a mobile phone, or drive a mini OTG DC fan (refer Fig. 6). For emergency light, six 4-volt strips, with ten LEDs on each strip, are used in parallel. Switch S1 is used to switch the emergency light on/off.

XL6009 booster module

When the power bank's output is connected to a load such as an emergency light, its battery voltage slowly starts dropping. To maintain a constant voltage level for charging a mobile phone or driving a mini fan, an XL6009 DC-DC step-up converter module is used (via switch S2). It is basically an adjustable power booster module that raises the power bank's output (low battery voltage) from 4.1 volts to 5 volts.

Using XL6009's inbuilt trimmer,



Fig. 7: Female USB sockets



Fig. 8: Author's finished prototype from outside

its output voltage can be increased to 5.5 volts for quick charging of a smartphone. But this voltage needs to be limited to 5.5 volts and should not be increased beyond this level for safety of the connected device. Recommended set point of output voltage is 5V.

The output is available through a standard female USB socket for charging mobile phones using a USB cable. This mobile charging output is available at CON3 in the circuit.

Another micro USB female socket is used at CON4 for driving the OTG mini fan. Switch S4 is used to switch off OTG fan. Typical female USB sockets available in the market are shown in Fig. 7.

A mini digital voltmeter is used to show the battery bank's voltage and boost converter's output voltage to

PARTS LIST

Semiconductors:

Board1
- TP4056 micro USB 5V, 1A 18650 Li-ion battery charging board with LED indicator

Board2
- DC-DC boost converter module XL6009

D1
- 1N4007 rectifier diode

Miscellaneous:

CON1, CON4
- Micro USB for TP4056 module and OTG fan

CON2
- 2-pin connector

CON3
- USB type A female socket for mobile charging

S1, S2, S4
- On/ off toggle switch

S3
- Single-pole double-throw (SPDT) switch

- 3,000mAh Li-ion 18650 rechargeable battery pack (4 nos.)

- 4V white LED strip with aluminium sheet

- OTG mini fan

- 0.71cm (0.28-inch), 2.5V to 30V range digital voltmeter

monitor the power bank's condition and mobile phone's charging state. A single-pole, double-throw (SPDT) switch (S3) is used to select between the power bank output and mobile charging output voltages.

Construction

A 17.78 × 10.16 × 3.81cm (7 × 4 × 1.5-inch) size plastic box can be used for housing the entire circuitry. It can accommodate all the components including Li-ion battery bank and LED lights.

The slots for switches and USB sockets can be provided on the enclosure as shown in Fig. 8. You may use a small knife or a poker to make holes in the plastic box for the sockets and switches and fix them using hot glue and double-sided tape.

A transparent acrylic sheet can be used on the front side for covering the LEDs of emergency light. Since aluminium sheet is present at the back of each LED strip, no separate heat-sink is required.

The author's finished prototype for the emergency light is shown in Fig. 8. **EFY**

Gautam Kumar Mandal is deputy general manager in Durgapur Steel Plant (SAIL). He did his M. Tech from NIT, Durgapur and B.E. from BE College, Shibpur (now IEST-Shibpur). He is an electronics hobbyist and a regular reader and contributor to EFY